



Comparative life cycle GHG emissions of single-use plastic cups and reusable cups for beverages

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ABSTRACT

Single-use plastic (SUP) products are widely used due to their convenience and low cost, with SUP cups being particularly prevalent in coffee shops and fast-food restaurants. This study estimates SUP cup consumption and evaluates their environmental impacts, focusing on global warming, through a cradle-to-grave life cycle assessment (LCA). Five scenarios (S1-S5) were set and compared with conventional SUP cups. Results indicate that 5.8 billion SUP cups were consumed in coffee shops and fast-food restaurants in South Korea in 2022, equating to 113 cups per person annually. The global warming potential (GWP) for the baseline scenario (S1) was 8.12 kgCO₂eq per functional unit, with SUP polymer production being the largest contributor. Scenario 5 (S5), involving a 25% substitution with reusable plastic cups and tumblers, achieved a 20% GWP reduction compared to S1. However, increased raw material use, energy demands, and cleaning processes for reusable cups led to higher burdens on land use and mineral extraction. It was found that reuse rate of reusable cups significantly influenced the results. This study highlights the effectiveness of deposit return scheme (DRS) and reusable cup policies, providing essential data to guide regulatory measures that mitigate SUP cup impacts and promote sustainable alternatives.

Keywords: Carbon footprint, Deposit return scheme (DRS), Life cycle assessment (LCA), Reusable cups, Single-use plastic (SUP) cups



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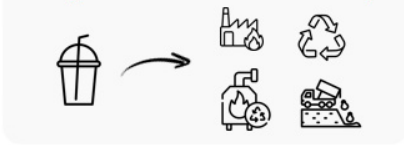
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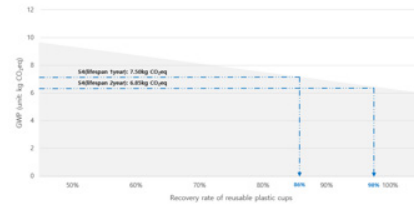
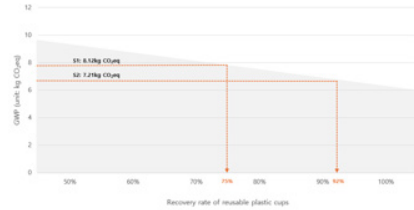
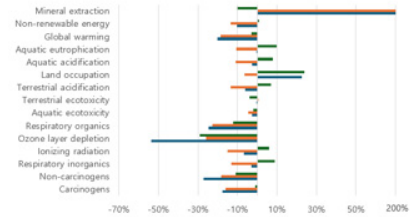
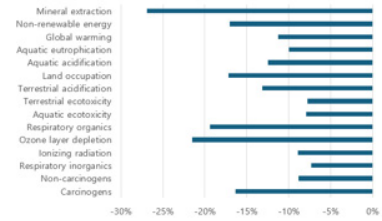
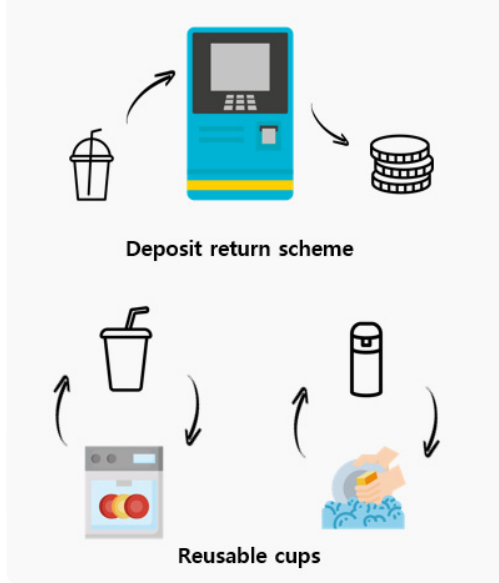
Graphical Abstract

Life Cycle Assessment of SUP cups



Environmental benefits of DRS &

Reusable cups for beverages



1. Introduction

The global demand for plastics is expected to surge from 4.5 billion tons in 2021 to nearly 1.2 billion tons by 2060 [1]. As of 2021, the packaging sector, which is a primary source of single-use plastics (SUP), represents the largest share of global plastic production, accounting for 44% [2]. SUP, which are intended to be used once (or for a very short period) before disposal, are predominantly used for food and product packaging, encompassing a variety of items such as plastic bags, cups, wrappers, water bottles, and containers [3]. Global production of SUP rose from 133 Mt in 2019 to approximately 139 Mt by 2021 [4]. This trend is anticipated to persist, with annual production potentially reaching 300 Mt by 2040, contributing significantly to the accumulation of plastic waste worldwide [5]. Much of SUP ends up with incineration, landfilling, and limited recycling after collection [6]. This contributes to environmental challenges, as plastic cups take an extensive period to decompose and may generate microplastics (MPs) that enter water bodies [7]. MPs ultimately enter the food chain and create severe health issues for humans and other species in the environment [7,8]. It should be noted that in developing countries,

where the infrastructure for solid waste collection and treatment is often lacking, large amounts of SUPs end up in open environments (rivers, streams, estuaries, and oceans) through leakage and uncollected pathways [9]. In addition, incineration of plastic cups further can exacerbate environmental issues by emitting greenhouse gases (GHGs) and toxic pollutants [7]. Recent studies indicate that global plastic pollution further exacerbates climate change, attributing to approximately 2.0 Gt CO₂ emissions in 2015 throughout the plastics life cycle [10].

Over the growing concerns of SUP overconsumption and their potential impacts, the European Union (EU) has implemented regulations for reduction of SUP. The EU SUP Directive aims to prevent and reduce the impacts of certain plastic packaging and products through market restrictions, consumption reduction measures, separate collection and design requirements, labelling requirements, extended producer responsibility, and information systems and reporting requirements [11]. In addition, the EU Packaging and Packaging Waste Regulation sets out a comprehensive framework to reduce packaging waste and promote a circular economy for packaging. The regulation targets a reduction in plastic packaging waste in 2018 by 5% by 2030, 10% by 2035, and 15% by 2040.

It also sets recycling targets for plastic packaging at 50% by 2025 and 55% by 2030. Furthermore, the EU requires final distributors in the HORECA sector (hotel, restaurant, café) by offering beverages or ready-prepared food in take-away packaging (of all materials) to provide consumers with refill and reuse options at no additional cost [12].

In South Korea, regulations on single-use or single-use items including SUP have progressively strengthened, starting with the ban on plastic bags in large retail stores in 2019 [13]. In response to the surge in plastic waste observed during the COVID-19 pandemic, the country further strengthened its restrictions on SUPs. On April 1, 2022, the use of SUP cups was banned in cafes, and from November 24, 2022, the prohibition extended to include single-use paper cups, plastic straws, and stirrers in restaurants and cafeterias, as well as plastic bags in convenience stores and bakeries [14]. According to the Korea Ministry of Environment (Korea MOE), the majority of SUP cups in Korea are disposed of through landfilling or incineration, with a very low recycling rate of around 5%. Most of these cups are not properly separated for recycling, whether they end up discarded on streets or mixed in disposal bags [15]. Thus, the country reintroduced the deposit return scheme (DRS) for single-use paper and plastics cups to prevent littering and boost their recycling after its abolishment in 2008. Under this system, food and beverage franchise stores are required to charge a deposit fee of KRW 300 (approximately USD 0.20), refundable upon the return of the single-use cup [16]. The pilot program has been conducted in two major cities (Sejong and Jeju) since 2022. Further, Korea has promoted reusable alternatives in packaging through initiatives such as “environmental label certification” for reusable container services [17] and hygiene guidelines standardizing production, management, and washing processes [18].

Along with the efforts on the reduction of consumption of SUP cups by policy measures and interventions, the scientific community have conducted life cycle assessment (LCA) to examine their environmental impacts by comparing SUP cups with reusable alternatives [19,20,21,22]. The LCA is one of the most recognized and widely used tool for assessing a product's potential environmental sustainability in supporting decision making and policymaking. Moretti et al. (2021) presented a comparative LCA of polylactic acid (PLA), polyethylene terephthalate (PET), and polypropylene (PP) cups. They found that the environmental impacts of PLA were lower than those of PET across all impact categories, while PP was favored over PLA in most categories, except for fossil fuel depletion and climate change [19]. Sadeleer et al. (2022) and Cottafava et al. (2021) analyzed the environmental impacts of single-use cups versus reusable cups [20,21], whereas Changwichan et al. (2020) compared SUP cups made from PLA, PET, and PP with reusable stainless-steel cups [22]. These studies concluded that reusable cups, including stainless steel options, are environmentally superior to SUP cups in beverage packaging applications. Additionally, Foteinis et al. (2020) found that paper cup consumption in the UK has a significant carbon footprint, which recycling could reduce by up to 40%, while switching to reusable cups achieves a threefold reduction in emissions, highlighting the environmental benefits of sustainable alternatives [23]. The outcomes of these studies varied depending on system boundary

conditions, such as product weight, number of reuse cycles, waste management practices, and the energy mix used in production and recycling processes. Recently, studies have comprehensively examined the environmental impacts of food and beverage packaging while also considering socio-economic indicators [24, 25, 26]. Caspers et al. (2023) and Corona et al. (2024) emphasized the importance of incorporating consumer behavior into sustainability assessments of packaging systems, particularly regarding reuse rates and disposal practices [24, 25]. Furthermore, Russo et al. (2023) proposed an expanded Life Cycle Sustainability Assessment (LCSA) framework by integrating additional environmental and socio-economic indicators, including material persistence, pollution, costs, and job creation [26]. However, there are still very limited LCA studies regarding carbon footprint of SUP cups so far. To the best of our knowledge, no previous LCA study in South Korea has specifically focused on SUP cups. Also, with the implementation of DRS and the substitution of reusable cups, including reusable plastic cups and tumblers as potential solutions to address the environmental challenges of SUP cups, it is essential to evaluate the environmental impacts caused by different consumption systems. Therefore, it is necessary to identify the major sources and contributing factors to the environmental impacts of SUP cups and assess the potential environmental benefits of DRS and reusable cups in the future.

To bridge these knowledge gaps, this study assesses the environmental impacts of SUP cups with a focus on carbon footprint and examines the potential benefits of transitioning away from conventional beverage cup consumption using the LCA methodology. Specifically, it estimates the consumption of SUP cups in South Korea in 2022 and evaluates the impact reduction through the implementation of a DRS-enhancing recycling rates and the shift to reusable alternatives by means of scenario analysis. The scientific findings of this study can be used for developing proper management plans for SUP cups and help identify priority of targeted policy interventions. Furthermore, the results can support the evaluation of policies related to SUP cups, contributing to the efforts to establish a sustainable and circular plastic society in the country.

2. Methodology

2.1. Data Acquisition

In this study, we estimated the annual consumption of SUP cups in South Korea in the year of 2022, focusing on coffee shops and fast-food restaurants, which are the most common users of single-use cups [27]. Statistical data of single-use cups were collected from the internal data of Korea Resource Circulation Agency (KORA), and population, number of relevant industries in South Korea were gathered from the Statistics Korea (KOSTAT) [28,29]. In addition, annual consumption rate of single-use cups (paper and plastic) per store was acquired from a research report of Daejeon Green Environment Center (DJGEC) in South Korea [30].

A comparative LCA of SUP cups and reusable cups, including reusable plastic cups and tumblers, was conducted using SimaPro version 9.4 software. The collection rate of the reusable plastic

cups was collected from the business report of a reusable packaging company in South Korea and the number of reuses before disposal was calculated referring to the Sadeleer et al. (2022) [20]. In addition, an interview was conducted with a reusable packaging company to collect information on the transportation of reusable plastic cups to the cleaning site. In the case of tumbler, the average lifespan was assumed based on previous studies on metal (stainless steel, aluminum) cups [22,31]. The amount of water, detergent, and electricity consumed in the cleaning process of reusable plastic cups and tumblers were set based on data from Korea Consumer Agency (KCA) [32,33]. Table 1 presents the detailed data collection for SUP cups, reusable plastic cups, and tumblers in this study.

2.2. Estimation of the consumption of SUP cups

The usage rate of SUP cups is defined as the quantity consumed per capita per year, in the unit of g/person/year or cups/person/year. In this study, the usage rate of SUP cups was calculated using Eq. (1) and Eq. (2).

$$\text{Plastic cup consumption(cups/person/year)} = (N \times AU \times P)/C \quad (1)$$

$$\text{Plastic cup consumption(g/person/year)} = (N \times AU \times P) \times W/C \quad (2)$$

where N is the Number of coffee shops and fast-food restaurants; AU is the annual usage of single-use cups per store; P is the percentage of plastic cups; W is the weight of SUP cup; C is the population.

2.3. Environmental impact of SUP cups and reusable cups

This study adheres to the LCA methodology and guidelines of the International Organization for Standardization (ISO) 14040-14044:2006. The LCA process was implemented in sequen-

tial steps: (a) defining the goal and scope; (b) conducting data collection and inventory analysis; (c) evaluating environmental impacts; and (d) interpreting the results [34,35].

2.3.1. Goal and scope definition

The aim of this study is to assess the environmental impacts of different beverage cups used, focusing on global warming. The potential reduction of environmental impacts by implementing DRS for SUP cups and replacing them with reusable cups and tumblers was studied. The consumption of SUP cups derived from this study was used as a functional unit (FU), defined as 113 cups per year, each with a capacity of 473ml, used for containing cold drinks. The volume of 473ml was selected as it is one of the most common volumes offered on the market in South Korea. In this study, both body and lid of cups were considered, and the weight of each cup was determined through direct measurement. Weight and material of cups are summarized in Table S1.

Based on the market survey, the ratio of PET and PP materials for SUP cups was set at 90%:10%. Furthermore, the material of the lid and body of SUP cups is assumed to be identical. The weight of the 473mL SUP cups was assumed to be 16g for PET and lighter when made from PP (12g). For the same volume (473ml), the weights of a reusable plastic cup and a tumbler are assumed to be 48g and 387g, respectively. In the case of the reusable plastic cup, the lid is considered single-use and made from PET. The system boundary of this study is set to be “cradle-to-grave”, considering from raw material production to disposal. In particular, the effect of reuse was analyzed by comparing the flow of SUP cups, which are disposed of after one use, and reusable plastic cups and tumblers, which can be reused multiple times. Fig. 1 shows the generic system boundaries of each cup, with the assumption that the use phase has negligible impacts for SUP cups.

Table 1. Data collection of SUP cups, reusable plastic cups, and tumblers.

Category	Data	References
Consumption of SUP cups		
Number of coffee shops and fast-food restaurants	145,092 stores	Statistics Korea (2024) [29]
Annual usages of single-use cups per store	68,138 cups/store/year	DJGEC (2022) [30]
Percentage of paper and plastic cups	41%:59%	Internal data of KORA (2020)
Population	51,690,000 people	Statistics Korea (2024) [28]
Environmental impacts		
Reusable plastic cups		
Transportation	32km	Interview with a Reusable packaging company
Collection rate	80%	Business report of a Reusable packaging company
Times of reuse	5 uses	Sadeleer et al (2022) [20]
Dishwashing	0.35L (tap water) 0.375mL (detergent) 30.875Wh (electricity)	Korea Consumer Agency (2021) [32]
Tumblers		
Life span	3 years	Changwichan et al (2020) [22]; Tamburini et al (2021) [31]
Hand-washing	1L (tap water) 2mL (detergent)	Korea Consumer Agency (2021) [33]

The disposal of tumbler was not considered in this study as they do not reach the end of their functional life (3 years) within one year. In accordance with the Korea MOE guidelines for environmental labeling certification [36], this study applied a 95% cumulative weight contribution cut-off for collecting data on product raw materials, which excludes rubber as a packing material for tumblers. Both reusable plastic cups and tumbler, the cleaning process was included in the use phase. Dishwashing and handwashing are assumed for reusable plastic cups and tumblers, respectively. While the cleaning of reusable plastic cups includes a seven-step process, this study only considered the cleaning process using a dishwasher. The round-trip transportation of reusable plastic cups to and from the cleaning plant was considered, however, the raw materials, transportation of the cups, and transportation to waste disposal facilities were excluded due to lack of data. Production and transportation of secondary and tertiary packaging were also not considered in all cup systems.

2.3.2. Life cycle inventory analysis

Foreground data, including material requirements, washing processes, and End-of-Life (EoL) pathways, were obtained from peer-reviewed literature, reports, and communications with a reusable plastic cup supplier. Background data for material production and process-related burdens throughout life cycle stages were extracted from the Ecoinvent v3 database. In this LCA, SUP cups were considered to be made from petrochemical PET and PP polymers, which are then thermoformed into cups. The extrusion of plastic sheets and the thermoforming process from Ecoinvent v3 was used. After the cups are used, it was assumed that they are managed the same way as municipal solid plastic waste in South Korea. In 2022, municipal solid plastic waste was treated with 16.4% mechanical recycling, 32.6% incineration, 38.2% energy recovery and 12.8% landfilling [37,38].

Reusable plastic cups were considered to be made from polypropylene (PP) and then subjected to the injection molding process. The injection molding data were obtained from the Ecoinvent database and adapted to the FU. For the PET cap production, the process extrusion of plastic sheets and thermoforming was used. The collected cups then needed to be delivered to the washing facility, and a cleaned cup was assumed to be delivered for customer use. It was assumed, there were centralized washing facility 32km (round trip) from the selling points. The transportation of the cups for washing was assumed to be done by a lorry with a capacity of 3.5 to 7.5 metric tons, which meets the Euro 6 emissions standard, as shown in Table S3. It was assumed that reusable plastic cups would be washed in a dishwasher every time after their use. For dishwashing, water consumption, detergent use and energy demand were allocated to reusable plastic cups on the assumption that the capacity of the industrial washing machine would be about 40 dishes/one cycle. Thus, during a single wash cycle, one cup was allocated 0.35L of water, 0.375mL of detergent and 30.875Wh of electricity, based on data derived from the Korea Consumer Agency's performance and quality comparison analysis of dishwashers [32]. The detergent used in the process is alkylbenzene sulfonate, assumed based on typical ingredients found in dishwashing detergent. Considering that reusable plastic cups are made and disposed of as a single material, high recycling rate could be applied. Specifically, 80% of reusable plastic cups were assumed to be mechanically recycled, while the remaining were disposed of through incineration (10%) or energy recovery (10%) due to contamination. In the case of tumblers, they were considered to be made from stainless steel, and the metalworking process from Ecoinvent v3 was used. Powder coating and enameling processes were also adapted for body production. For the PP cap production, the injection molding process from the Ecoinvent database was used. Similar to reusable plastic cups, tumblers were

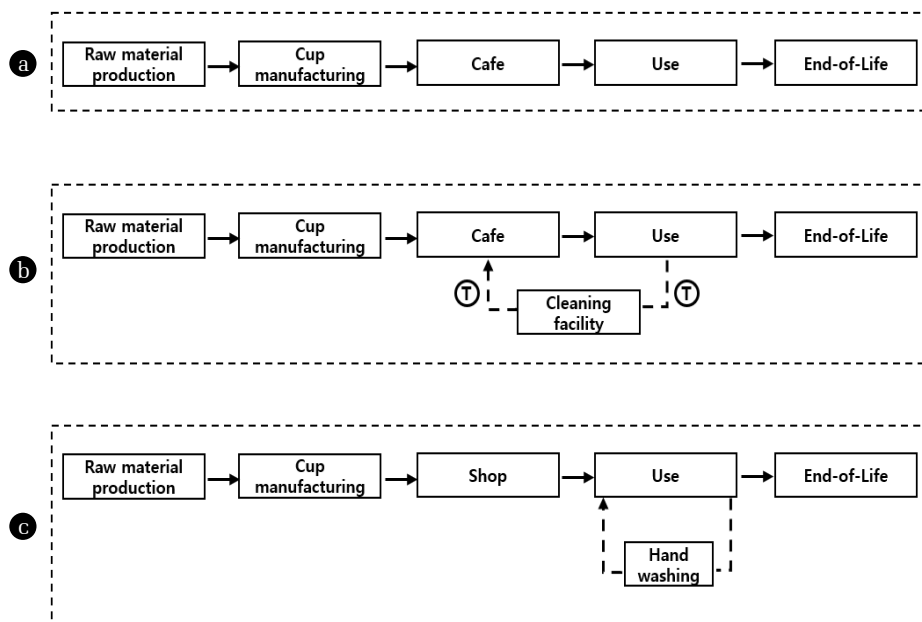


Fig. 1. System boundary of beverage cups: (a) SUP cup, (b) reusable plastic cup, (c) tumbler.

assumed to be hand-washed after each use. Based on the quality comparison test of kitchen detergents reported by the Korea Consumer Agency [33], it was assumed that 1 L of water and 2 mL of detergent were consumed per cup during a single handwashing cycle in households. The same detergent (alkyl benzene sulfonate) was assumed for tumblers as for reusable plastic cups. A summary of the main life cycle inventory (LCI) inputs for material and processes across various life cycle stages for the three cup types is provided in Table S1, Table S2, and Table S3.

2.4. Scenario analysis

Assumptions and description in each scenario are summarized in Table 2, based on the end-of-life options for SUP cups and the potential uses of reusable plastic cups and tumblers. Scenario 1 (S1) represents the existing scenario, assuming the daily use of 113 SUP cups. In Scenario 2 (S2), we considered the case where the rate of material recycling and energy recovery of SUP cups increase due to the implementation of the DRS for single-use cup. For this purpose, we referenced the recovery rate of the DRS piloted in Sejong City in 2022 (approximately 40%) [39] and the methodology from prior research [40]. It was assumed that 90% of the collected single-use cups are recycled (36%). Additionally, the recovery rate is assumed to reach 78%, reflecting the performance observed in Jeju City [39], with the remaining portion of the collected cups that are not materially recycled being processed through energy recovery. The uncollected 22% of SUP cups are assumed to be entirely incinerated, as landfilling is excluded due to the ban on municipal waste landfilling without pretreatment, which will take effect in 2026 for the three metro cities and in 2030 for all other cities [41]. Scenarios 3 and 4 assume that 25% of existing SUP cups are replaced by reusable plastic cups and tumblers, respectively. Finally, Scenario 5 (S5) assumes a mixed-use model, with SUP cups, reusable plastic cups, and tumblers used in proportions of 50%, 25%, and 25%, respectively.

For reusable plastic cups, a collection rate of 80% was considered, based on data from a reusable plastic cup supplier. Accordingly, it is assumed that only 80% of the initial number of reusable plastic cups enters the reuse circulation system, while the 20% that is not recovered is disposed of. The average number of reuses and the initial number of cups produced were calculated using

Eq. (3) and Eq. (4) [20]. Therefore, in scenarios 3 and 5, it is assumed that 5.65 reusable plastic cups are required initially, with each cup being reused five times. A detailed breakdown of the reuse stages of the reusable plastic cups is provided in Fig. S1.

$$\text{Average number of reuses} = 1 / (1 - \text{collection rate}) \quad (3)$$

$$\text{Initial number of production} = \text{Serving number} / \text{Average number of reuse} \quad (4)$$

For tumblers, a lifespan of 3 years was assumed, based on previous studies [22,31]. Therefore, in scenarios 4 and 5, it was assumed that each tumbler is used a total of 28.25 times over its lifespan. Fig. S2 presents a detailed breakdown of the reuse stages for the tumblers.

Life cycle impact assessment (LCIA) was conducted using the IMPACT 2002+ V2.15 impact assessment method, as it provides a comprehensive characterization of environmental impacts and allows for a clear communication of results by combining midpoint indicators into a single endpoint indicator, which is commonly used in LCA studies [42]. LCA results are presented as fifteen impact categories: Carcinogens, Non-carcinogens, Respiratory inorganics, Ionizing radiation, Ozone layer depletion, Respiratory organics, Aquatic ecotoxicity, terrestrial ecotoxicity, Terrestrial acidification/nutritification, Land occupation, Aquatic acidification, Aquatic eutrophication, Global warming, Non-renewable energy, Mineral extraction.

3. Results and discussion

3.1. Results of SUP cup consumption analysis

This study employed a top-down approach to estimate the consumption of SUP cups in coffee shops and fast-food restaurants. In 2022, it was estimated that approximately 5.8 billion SUP cups were consumed nationwide, equivalent to 81,200 tons based on an average weight of 14 grams per cup. This volume accounts for nearly 1.5% of the total municipal plastic waste generated in 2022, which was around 5.26 million tons [38]. Furthermore, on a per capita basis, this is equivalent to 113 SUP cups annually, or two cups per week.

Table 2. Major assumptions in each scenario analysis in this study.

Scenarios	Assumptions	End-of-life scenario
S1 (BAU)	113 SUP cups	Recycling 16.4%, Incineration 32.6%, Energy recovery 38.2%, Landfilling 12.8%
S2 (Implementation of DRS)	113 SUP cups	Recycling 36%, Incineration 22%, Energy recovery 42%
S3 (Substitution of reusable plastic cups)	84.75 SUP cups, 28.25 reusable plastic cups	- SUP cups are disposed of the same as S1 (BAU) - For reusable plastic cups, recycling 80%, incineration 10%, energy recovery 10%
S4 (Substitution of tumblers)	84.75 SUP cups, 28.25 tumblers	- SUP cups are disposed of the same as S1 (BAU) - Disposal of tumbler was not considered
S5 (Substitution of reusable plastic cups and tumblers)	56.5 SUP cups, 28.25 reusable plastic cups, 28.25 tumblers	- SUP cups are disposed of the same as S1 (BAU) - For reusable plastic cups, recycling 80%, incineration 10%, energy recovery 10% - Disposal of tumbler was not considered

According to the Korea MOE, approximately 420 million single-use cups (including paper and plastic) were used in 2007 in coffee shops, bakeries, and fast-food restaurants, with an estimated 252 million SUP cups based on a distribution of 40% paper and 60% plastic cups [43]. By 2019, Korea MOE estimated that annual consumption of single-use cups had risen to approximately 8.4 billion, comprising 3.7 billion paper cups and 4.7 billion plastic cups [15]. Further, Greenpeace Korea reported that around 3.3 billion SUP cups were consumed in 2017 in coffee shops and fast-food restaurants [44]. During the COVID-19 pandemic, they found that SUP cup consumption increased to approximately 5.3 billion in 2020 [45]. In South Korea, the consumption of SUP cups have increased significantly over the last decade, likely driven by the proliferation of coffee shops. This trend was further exacerbated during the COVID-19 pandemic due to the rise in take-aways and food deliveries, as well as the government's temporary relaxation of restrictions on single-use products.

With the global rise in single-use cup usage, many studies have examined their consumption patterns. Table 3 presents a comparative analysis of single-use cup consumption in South Korea, Japan [46,47], and Germany [48,49]. In Japan, Itochu Pulp & Paper Corporation reported a total consumption of 3.9 billion single-use cups in 2020 [46]. Additionally, Greenpeace Japan found that nine major coffee chains, representing approximately 50% of the country's coffee market, accounted for 379.5 million cups [47]. In Germany, estimates by GVM [48] and the Berlin Senate Department [49] indicated that single-use cup consumption nearly doubled from 1.5 billion in 2017 to 2.8 billion in 2024.

SUP cup consumption rates vary by country due to several factors such as consumers' behavior, social culture, policy measures, and environmental awareness, limiting direct cross-country comparisons. However, per capita usage of SUP cups in Korea is relatively high compared to other countries. In line with the global trend of reducing plastic waste, it is necessary to curb the consumption of SUP cups, support widespread use of reusable cups, and enhance resource circularity through policies such as banning the free provision of SUP cups, implementing DRS for single-use cups, and offering incentives to customers who bring personal cups.

3.2. Life cycle impact assessment results of SUP cups and reusable cups

3.2.1. Baseline scenario (Scenario 1, S1)

The overall breakdown for the environmental impacts of 113 SUP cups by LCA is presented in Fig. 2(a). Polymer production repre-

sents the major environmental burden for 13 impact categories, causing over 40% of the impacts. For two categories, Ionizing Radiation and Terrestrial Ecotoxicity, the cup production phase was the dominant contributor, accounting for 60.67% and 58.54% of the impacts, respectively. The Ionizing Radiation category measures human exposure to radiation over time, mainly resulting from nuclear energy generation [50]. In South Korea, nuclear power constitutes 30.7% of the electricity mix, with coal at 31.4%, LNG at 26.8%, and renewables at 9.6% as of 2023 [51]. Therefore, energy-intensive processes, such as plastic sheet production and thermoforming, which heavily depend on this energy mix, emit significant levels of ionizing radiation. Similarly, cup production has a substantial impact on Terrestrial Ecotoxicity, primarily due to the release of heavy metals like zinc (Zn), aluminum (Al), and copper (Cu). These metals can be accumulated to the soil, disrupting soil-based ecosystems, including plants and microorganisms, and may bioaccumulate in the human body through the food chain [52]. To produce 113 SUP cups, zinc is the largest contributor to Terrestrial Ecotoxicity, accounting for 43.3 kg TEG soil, followed by aluminum and copper at 36.5 kg TEG soil and 28.5 kg TEG soil, respectively.

The global warming impact was found to be about 8.12 kg CO₂ eq/f.u in this study. Fig. 2(b) shows the detailed contributions to global warming impact. Polymer production is responsible for 54.6% of global warming impact, with virgin PET and virgin PP production contributing 51.2% and 3.4%, respectively. Manufacturing SUP cups is the second major source of impact, accounting for 23.4%, followed by disposal (22%). For cup manufacturing, the primary source of GHG emissions is the high electricity demand required for plastic sheet extrusion and thermoforming. In the disposal phase, incineration (14.82%) and energy recovery (14.14%) are major contributors to global warming due to the substantial carbon emissions released during plastic incineration. By contrast, landfilling has a minor impact on carbon emissions (0.13%) at the end-of-life (EoL) stage, while material recycling provides an environmental benefit of -7.09%, offsetting emissions through resource recovery.

3.2.2. Environmental impact reduction through DRS (Scenario 2, S2)

The implementation of a DRS for SUP cups resulted in reductions across all impact categories, with the most significant decrease observed in the Mineral Extraction category (-27%). This improvement is largely attributed to reduced landfilling, increased recycling, and enhanced energy recovery. These measures decrease reliance on virgin material extraction, thereby conserving resources and minimizing ecological impacts from raw material depletion.

Table 3. Comparison of annual cup consumption between South Korea, Japan, and Germany.

Country	Annual consumption	Year	Scope	Reference
South Korea	5.8 billion SUP cups	2022	Nationwide	This study
Japan	3.9 billion single-use cups	2020	Nationwide	Itochu Pulp & Paper Corporation, 2020 [46]
	0.4 billion single-use cups	2020	9 major Japanese coffee chains	Greenpeace Japan, 2022 [47]
	1.5 billion single-use cups	2017	Nationwide	NABU, 2018 [48]
Germany	2.8 billion single-use cups	2024	Nationwide	Senatsverwaltung für Mobilität, Verkehr, Klimaschutz und Umwelt, 2024 [49]

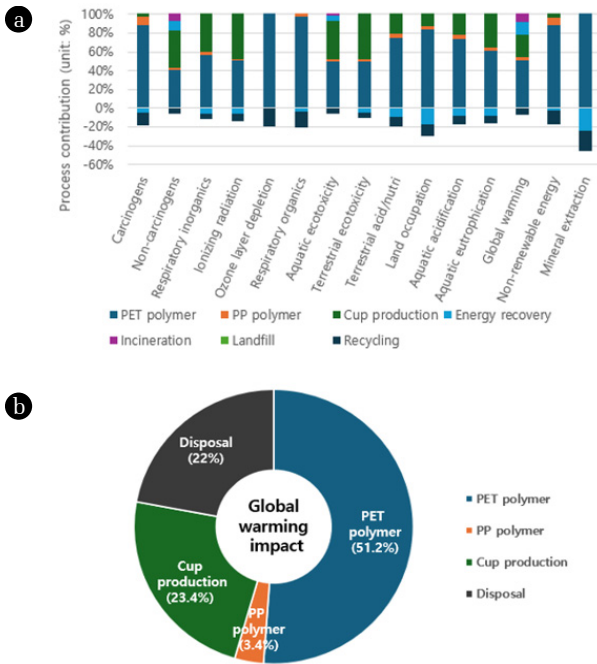


Fig. 2. (a) Breakdown (%) of the environmental impacts, (b) Process contributions to the global warming impact of S1.

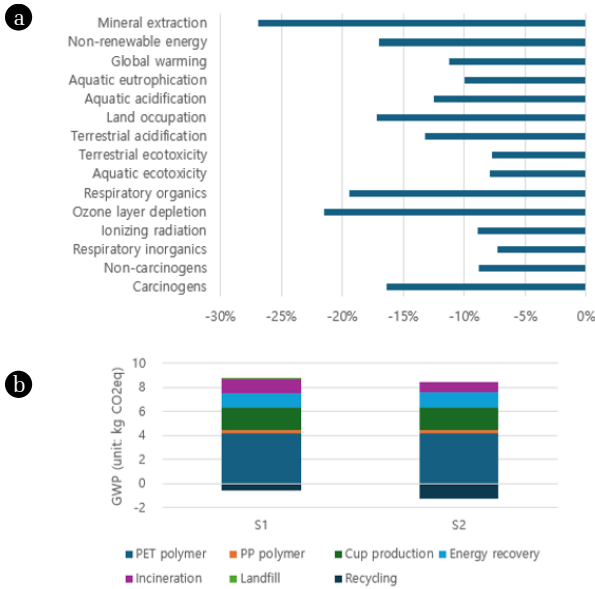


Fig. 3. (a) Relative comparison of the environmental impacts of S2 with S1, (b) Process contributions to the global warming impacts between S1 and S2.

The global warming impact was reduced to 7.21 kg CO₂ eq/f.u., which is an 11.3% decrease, compared to that of the baseline scenario (S1). Although increased energy recovery at the disposal stage slightly raised emissions (+0.14 kg CO₂ eq), this was offset by lower emissions from increased material recycling (-0.66 kg CO₂ eq) and reduced incineration (-0.39 kg CO₂ eq). Additionally,

the elimination of landfilling provided an extra reduction of 0.01 kg CO₂ eq.

3.2.3. Environmental benefits of Reusable Cups: Reusable plastic cups and Tumblers (Scenario 3, Scenario 4, Scenario 5)

Fig. 4(a) and Fig. 4(b) show the relative comparison of environmental impacts of reusable plastic cups, tumblers with those of SUP cups. Compared to the baseline scenario (S1), reusable plastic cups by replacing SUP cups at a 25% substitution rate (S3) created environmental benefits in terms of 8 impact categories. In the remaining seven impact categories, S3 showed higher impacts with the largest increase seen in land occupation (+24%). This increase is primarily attributed to the greater material demands of reusable plastic cups, which require three to four times more polymer than SUP cups to ensure durability. Moreover, their life cycle involves additional processes such as transportation to washing facility and energy-intensive cleaning, which increase resource and energy consumption. These factors, along with the infrastructure needed for cleaning and maintenance, contribute to higher environmental impacts, including greater land occupation. Using tumblers by replacing SUP cups at a 25% substitution rate (S4) also provided environmental benefits across almost all impact categories except for terrestrial ecotoxicity and mineral extraction. The mineral extraction impact in S4 is approximately three times higher than in S1, primarily due to tumbler manufacturing. While tumblers offer environmental advantages through extended use, their manufacturing process is resource-intensive, mainly due to the primary use of stainless steel. The production of stainless steel requires extracting metallic resources such as iron ore, chromium, and nickel [53], significantly amplifying the environmental impact in the mineral extraction phase. When replacing SUP cups with a combination of reusable plastic cups and tumblers at substitution rates of 25% each (S5), environmental benefits were observed in most impact categories, except for land occupation and mineral extraction. Notably, ozone layer depletion decreased by 53% compared to S1, primarily due to the reduction in virgin plastic production. Since chlorofluorocarbons (CFCs) and halogenated chlorofluorocarbons (HCFCs) are key substances associated with petroleum refining for plastic resin feedstock, their reduction significantly mitigates ozone depletion [54].

Compared to S1, both S3 and S4 showed more favorable outcomes for climate change, by offering 3% and 18% lower GHG emissions, respectively. The global warming impact for S3 was found to be about 7.88 kg CO₂ eq/f.u., representing a 3.0% decrease (-0.24 kg CO₂ eq) when compared to that of S1. The production of virgin PP (43%) and the manufacturing of cups through injection molding (28%) constitute the primary environmental burdens in terms of global warming impact throughout the cradle-to-grave life cycle of reusable plastic cups. Dishwasher electricity consumption for cleaning reusable plastic cups was the third largest contributor, contributing 18% of emissions. This high global warming impact of reusable plastic cup production is largely due to the intensive process, which requires larger plastic quantities and operates at elevated pressures and temperatures, driving up energy demand. The global warming impact of S4 was estimated to be at 6.70 kg CO₂ eq/f.u., representing a reduction of approximately 1.42 kg CO₂ eq (-18%) compared to the S1. Within the tumbler's

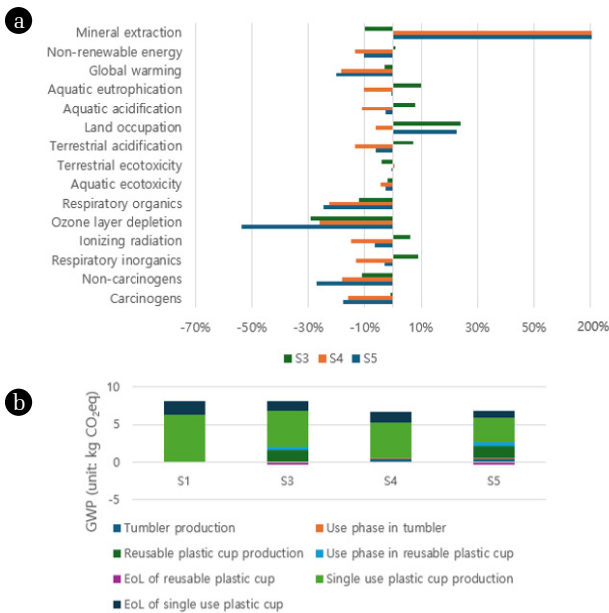


Fig. 4. (a) Relative comparison of the environmental impacts of S3-S5 with S1, (b) Process contributions to the global warming impacts between S1, S3, S4, and S5.

life cycle, production accounts for the largest share of the global warming impact (82%), primarily due to energy-intensive metal-working and molding processes that operate at high temperatures. Additionally, dishwashing detergents contribute 13% to the global warming impact, largely because the petroleum-based surfactants (e.g., alkylbenzene sulfonates, ABS) are used and produced through high-temperature processes that also emit GHGs [55].

Scenario 5 (S5), which assumes replacing 25% of SUP cups with reusable plastic cups and another 25% with tumblers, achieved the highest carbon reduction among all alternative scenarios. The global warming impact was reduced to about 6.49 kg CO₂ eq/f.u., representing a reduction of 20% (-1.63 kg CO₂ eq) when compared to the S1. In terms of process contributions, the production of SUP cups, including polymer production, accounted for 49% of total carbon emissions, while the production of reusable plastic cups contributed 25%. The disposal of SUP cups followed, contributing 14%. Other processes, including the transport and cleaning of reusable plastic cups at take-back facilities (8%), tumbler production (7%), tumbler cleaning (2%), and the disposal of reusable plastic cups (-4%), also contributed to the overall carbon footprint.

3.2.4. Carbon Footprint Analysis by Scenario

The study evaluated the carbon footprint of implementing a DRS for SUP cups and partially replacing them with reusable plastic cups or tumblers by five different scenarios (S1~S5). The results showed that increasing the recycling rate of SUP cups (S2) achieved greater carbon reductions than substituting 25% of SUP cups with tumblers (S3). Notably, the environmental benefits intensified as the substitution rate of SUP cups with tumblers increased, resulting in proportional carbon reductions. Annual carbon emissions for

each scenario (S1~S5) are summarized in Table S5. Under the baseline scenario (S1), with 113 SUP cups consumed per person annually in South Korea, total carbon emissions were estimated at 419,723 tons CO₂ eq. Increasing the recycling rate through the DRS is anticipated to reduce the emissions by 47,038 tons CO₂ eq. Substituting 25% of SUP cups with reusable plastic cups could lower emissions by 12,406 tons CO₂ eq annually, while replacing 25% with tumblers could achieve reductions of 77,050 tons CO₂ eq per year. At a substitution rate of 50%, annual carbon savings are projected to exceed 80,000 tons.

For reusable plastic cups, GHG emissions vary significantly with the recovery rate. Fig. 5(a) compares carbon emissions across different recovery rates for reusable plastic cups. When the recovery rate decreases from 80% to 70%, GHG emissions in S3 increase by approximately 0.53 kg CO₂ eq (+6.7%), surpassing the GHG emissions of S1. Conversely, increasing the recovery rate to 90% reduces GHG emissions by about 0.60 kg CO₂ eq (-7.6%). The analysis identifies a 75% recovery rate as the break-even point for reusable plastic cups (S3) in comparison to SUP cups (S1) in terms of global warming impact. This threshold suggests that when the recovery rate of reusable plastic cups exceeds 75%, they offer a lower carbon footprint than that of SUP cups. Furthermore, to match the GHG emissions of S2 (DRS), the recovery rate of reusable plastic cups would need to exceed 92%. The lifespan of tumblers also affects their global warming impact as presented in Fig. 5(b). Reducing the tumbler lifespan from three years to two years increases GHG emissions in S4 by approximately 0.2 kg CO₂ eq (+3%) and further reducing it to one year raises emissions by around 0.9 kg CO₂ eq (+13%). With a three-year lifespan, tumblers consistently exhibit a lower global warming impact than reusable plastic cups, regardless of the recovery rate of the cups. However, when the tumbler lifespan is reduced to two years, tum

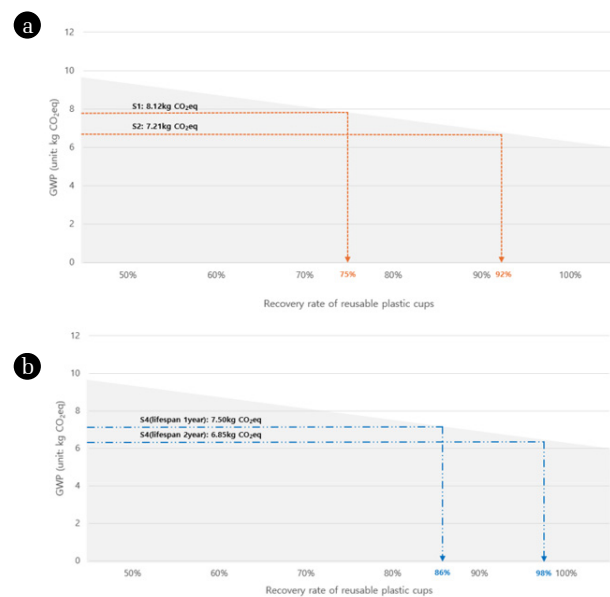


Fig. 5. Changes in carbon emissions based on the (a) recovery rate of reusable plastic cups (S3) and (b) lifespans of tumblers (S4). Shade area represents S3.

blers become less favorable in the global warming category once the recovery rate of reusable plastic cups exceeds 98%. Similarly, with a one-year lifespan, tumblers become less favorable when the recovery rate of reusable plastic cups exceeds 86%. Notably, when the tumbler lifespan is reduced to one year, GHG emissions surpass those of the SUP cup DRS (S2).

3.3. Sensitivity Analysis

Sensitivity analysis seeks to determine the level of sensitivity the individual input parameters have on the LCA results (the impact category measurements). This is achieved by conducting a perturbation analysis where each of the input parameters is varied within a range of 10% of its original value while maintaining the rest of the input parameters unaffected. In this study, sensitivity analysis was performed in regard to the collection distance and the amount of electricity, detergent, and tap water consumed for reusable plastic cups, focusing on the impact to global warming. As a result, a variation between -0.01% and +0.46% is observed in the cradle-to-grave results (Table S6). It showed that the global warming impact values are less sensitive to the collection distance of reusable plastic cups and electricity/detergent/tap water consumption at the washing facility.

3.4. Comparison with Previous Studies

In this study, SUP cups, reusable plastic cups, and tumblers emitted approximately 72 g CO₂eq, 63 g CO₂eq, and 19 g CO₂eq, respectively, per use. When compared with the results of previous studies (Table S7), GHG emissions of SUP cups were found to be similar, although discrepancies arose due to differences in material composition and disposal scenarios. However, reusable plastic cups and tumblers showed relatively higher GHG emissions. For reusable plastic cups, this difference is attributed to the assumption in this study of producing five cups and using each an average of 5.65 times. By contrast, previous studies assumed a single reusable plastic cup was used 28.25 times (with a recovery rate of 97%). When a LCA was conducted under these assumptions, the GHG emissions from one use of a reusable plastic cup decreased to approximately 26 g CO₂eq, closely aligning with prior research results. For tumblers, GHG emissions varied depending on material differences between the body and lid, as well as the presence and method of washing. This study included additional processes, such as metal fabrication and finishing (e.g., coating), which likely increased electricity consumption, resulting in higher GHG emissions compared to previous studies.

Greenpeace East Asia showed that as the number of uses of reusable plastic cups increased to 20, 40, and 60 times per year, GHG emissions decreased by 37%, 42%, and 44%, respectively [56]. Similarly, studies by Changwichan *et al.* (2020) [22] and Tamburini *et al.* (2021) [31] revealed significant reductions in environmental impact compared to the SUP option when the long-term use of tumblers was considered. These findings consistently demonstrate that reusable cups have substantial potential environmental benefits in replacing the continuous consumption of SUP cups, with greater reductions in GHG emissions achieved through repeated reuse.

3.5. Environmental Perspectives and Implications

This study showed that the production phase, particularly polymer manufacturing, dominates the environmental impact of SUP cups, accounting for over 75% of the global warming potential. Therefore, reducing the overall use of SUP products by promoting reusable takeaway cups is a critical step. Encouraging the adoption of reusable systems can significantly decrease the demand for virgin materials for SUP cup manufacturing, thereby lowering the associated upstream emissions.

At present, a large portion of used SUP cups are poorly managed and typically treated through landfilling or incineration processes that exacerbate environmental burdens and hinder recycling efforts. Securing used plastic cups through the DRS plays a pivotal role in mitigating these issues by incentivizing consumers to return cups for appropriate recycling. By enhancing material recovery rates, the DRS reduces waste generation, diverts plastics from landfills and incinerators, and supports the principles of a circular economy. Furthermore, it ensures that high-quality materials re-enter the production cycle, reducing the dependence on virgin resources and lowering the environmental footprint of new products. Expanding the DRS to encompass broader sectors—such as high-volume coffee shops, fast-food chains, and other takeaway-heavy businesses—is crucial for maximizing its environmental benefits. Such expansion would significantly increase the capture rate of SUP cups, thereby promoting more sustainable consumption patterns across a wider audience. However, to further enhance the effectiveness of the DRS, improvements to the current system are necessary. In South Korea, the DRS is currently operated through a mobile application that tracks returned cups, but this approach has introduced some inconvenience and complexity for users. Simplifying and streamlining the return process is essential to make it more consumer friendly. By reducing these barriers, the system can boost consumer participation and further improve recycling outcomes, ultimately contributing to higher material recovery rates and more effective waste management.

Furthermore, this study shows that the environmental advantages of reusable alternatives are highly dependent on key factors such as recovery rates, the number of reuse cycles, and the product lifespan. Ensuring high recovery rates is crucial to minimizing waste and maximizing the environmental benefits of reusable items. Likewise, extending the lifespan of reusable cups—through durable design, proper maintenance, and responsible use—can reduce the overall carbon footprint per use. This highlights the importance of public awareness campaigns to educate consumers on the benefits of reuse and proper handling of reusable cups. Moreover, investments in infrastructure—such as centralized washing facilities powered by renewable energy and efficient collection systems—are critical to supporting the widespread adoption of reusable options. Efforts should also focus on optimizing the cleaning process by using eco-friendly detergents and water-efficient systems to minimize environmental impacts. By addressing these challenges, reusable cups can provide a more sustainable solution, contributing to a reduction in overall environmental burdens.

4. Conclusions

This study conducted a quantitative analysis of the number and weight of SUP cups used in South Korea in 2022, aiming to estimate their consumption. It also compared the environmental footprints of SUP cups, reusable plastic cups, and tumblers, focusing on global warming impact through a comparative LCA. To evaluate potential reductions in environmental impact, five scenarios were developed, including the implementation of a DRS for SUP cups and the increased adoption of reusable plastic cups and tumblers.

By analyzing the coffee shop and fast-food restaurants—major consumers of SUP cups—the study found that approximately 5.8 billion SUP cups were consumed in South Korea in 2022, equating to an average of 113 SUP cups per person annually. Consumption within domestic coffee shops and fast-food restaurants has continued to rise, representing an approximately 74% increase in 2022 compared to 2017, when the average was 65 cups per person. Notably, per capita consumption of SUP cups in South Korea is significantly higher compared to international figures.

The LCA estimated that the total carbon footprint associated with the annual use of 113 SUP cups per person (S1) is approximately 419,723 ton CO₂ eq. A process-specific contribution analysis revealed that 13 out of 15 impact categories had the highest contributions from stages involving the production of virgin PET and PP. For the remaining two categories—ionizing radiation and soil ecotoxicity—the environmental impact of the cup production process itself was particularly significant. The analysis indicates that implementing a DRS (S2), which enhances material recycling and energy recovery rates for SUP cups, could reduce GHG emissions by approximately 47,038 tons CO₂ eq annually. Environmental impacts across 14 additional categories were also reduced, with a notable 27% decrease in mineral extraction impacts. Replacing 25% of SUP cups with reusable plastic cups (S3) or tumblers (S4) was estimated to reduce annual carbon emissions by 12,406 tons CO₂ eq and 77,050 tons CO₂ eq, respectively. A further shift to a 50% replacement with reusable plastic cups (S5) could achieve annual carbon savings exceeding 80,000 tons CO₂ eq. The analysis of contributions to the global warming category by unit processes indicates that, for reusable plastic cups, the primary environmental impacts are associated with cup production (including polymer manufacturing) and electricity consumption during dishwasher use. For tumblers, the most significant impacts are linked to cup production and detergent use. Across the remaining 14 impact categories, a 25% adoption of reusable plastic cups resulted in higher environmental impacts than S1 in seven categories, with the most significant increase observed in the land occupation category (+24%). Similarly, a 25% adoption of tumblers led to increased impacts in two categories, with mineral extraction nearly tripling compared to S1.

To evaluate the reliability of the results, a sensitivity analysis was conducted on transportation, detergent, electricity, and water consumption data within the life cycle of reusable plastic cups. The analysis revealed that all four factors had minimal influence on the LCIA results, particularly in the global warming category. Finally, the study analyzed the changes in GHG emissions based on the recovery rate of reusable plastic cups and the lifespan

of tumblers. For reusable plastic cups, GHG emissions varied with the recovery rate; when the recovery rate exceeded 75%, these cups demonstrated a more favorable global warming profile than SUP cups (S1). Furthermore, a recovery rate of at least 92% was required for their performance to align with that of the single-use cup DRS scenario (S2). For tumblers, the global warming impact increased as their lifespan shortened. When the lifespan was reduced to one or two years, their environmental impact exceeded that of reusable plastic cups with recovery rates of 86% and 98%, respectively.

This study utilized the Ecoinvent v3 LCI database, which relies on literature-based data and, as a result, has limitations in accurately reflecting conditions specific to South Korea. To develop a more representative national database, it is essential to incorporate actual industry data or systematically institutionalized statistical data. Currently, South Korea lags behind other countries in the development and regular updating of LCI databases, underscoring the need for continuous database improvement and effective management. In addition, evaluating the sustainability of a DRS for SUP cups and promoting the adoption of reusable cups should extend beyond environmental impact assessments. It is equally important to consider life cycle costs (LCC) and social impacts from an economic perspective. Furthermore, to achieve a holistic LCA of beverage cups, the environmental impacts of ICT infrastructures supporting the operation of the DRS for SUP cups, as well as other secondary services, must be assessed. These considerations underscore the necessity for further research that integrates environmental, economic, and social dimensions to support comprehensive decision-making.

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Author Contributions

C.L.(master's student) wrote the manuscript, developed methodology and conducted LCA study with software; H.L.(master's student) collected LCI data with curation; Y.C.J. (Professor) developed conceptualization, methodology and revised and edited the manuscript with supervision; K.C.(research professor) reviewed and edited the manuscript along with supervision; All authors have read and agreed to the published version of the manuscript.

Conflict-of-Interests Statement

The authors declare that they have no conflict of interest.

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